Applied Statistics for Engineers and Scientists II

ASESII 24A02 — HCMUS Advanced Program

Project-Based Quiz 1

Ridge, Lasso, and Regularization for Neural Features

Group 01

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Authorship and Contribution Statement

We, the undersigned members of **Group 01**, confirm that every member is able to explain any part of this analysis when asked. All members participated in the design, implementation, and review of this report. Any AI-assisted work is documented in the AI Usage Log (Appendix A).

Table 1: Individual contributions and verification responsibilities.

Student	**ID**	**Main Contributions**	**Verification Performed**
Nguyễn Đình Thiên Lộc	24125093	P1 — project configuration, data preprocessing, exploratory data analysis, data leakage prevention, and drafting the Prediction Design section.	Verified the exclusion of linearly dependent predictors and confirmed no test-set information leaked into the standardisation pipeline.
Trần Lê Anh Tuấn	24125107	P2 — OLS baseline, Ridge estimation, cross-validation curves, coefficient paths, and conditioning diagnostics.	Checked the shared-fold OLS and Ridge results, both tuning rules, generated tables, and diagnostic figures.
Lê Minh Thuận	24125105	P2 — Lasso estimation, controlled model comparison, core-model declaration, and bootstrap stability analysis.	Confirmed the Lasso active sets, shared-fold errors, pre-holdout selection rule, and bootstrap summaries.
Nguyễn Bảo Minh Triết	24125047	P3 — Ridge normal-equation and conditioning derivations, Lasso subgradient conditions, and evidence connection.	Re-derived the stated results under the same `glmnet` scaling and checked them against the numerical eigenvalues.
Nguyễn Hồng Tấn Tài	24125078	P4–P5 — Elastic Net, fixed ReLU features, locked holdout evaluation, integration, abstract, and reproducibility record.	Verified the frozen feature map, training-only hidden-feature handling, locked declaration, final metrics, and report build.

Abstract

We predict Brozek body-fat percentage from 14 anthropometric measurements for 252 adult men. Outcome-derived variables were excluded, preprocessing used training-only statistics, and OLS, Ridge, Lasso, Elastic Net, and penalised models on 200 fixed random ReLU features were compared with one shared

five-fold split. Lasso at $\lambda_{\min} = 0.0954$ had the lowest training cross-validated RMSE among the original-predictor models (4.241), retained nine slopes, and was declared for deployment before the holdout responses were examined. On the 51-subject holdout set it remained best, with RMSE 4.251, MAE 3.524, and $R^2 = 0.683$. Elastic Net was close (RMSE 4.274), whereas the fixed ReLU expansions did not improve prediction. We therefore recommend the predeclared Lasso for this population, while treating its selected variables as predictive rather than causal evidence.

1 Prediction Design and Leakage Control

1.1 Target and Predictors

The `fat.csv` dataset contains $n = 252$ observations and 18 variables describing body composition and anthropometric measurements of adult males [Johnson, 1996]. The unit of observation is one adult male subject's body measurement record, and the objective is to predict that subject's Brozek body-fat percentage from measurements available without a body-fat calculation. The response variable is `brozek`, the body-fat percentage estimated via Brozek's equation.

After inspection, the following variables are **excluded** from the predictor set to prevent data leakage:

- `siri` — an alternative body-fat estimate; including it would constitute a near-perfect proxy for the response. Both Siri and Brozek equations are derived directly from body density, making them fundamentally the same measure.
- `density` — body density is the direct input to both the Brozek and Siri formulas (e.g., Brozek = $457/\text{density} - 414.2$). Using it makes prediction trivially deterministic.
- `free` — fat-free weight is algebraically derived from body fat and total weight (free = weight $\times$ $(1 - \text{body fat }\%/100)$), creating a circular dependency and data leakage.

This leaves $p = 14$ candidate predictors (age, weight, height, adipos, neck, chest, abdom, hip, thigh, knee, ankle, biceps, forearm, wrist).

Summary statistics. Table 2 reports the descriptive statistics for all retained predictors computed on the *training set only*.

1.2 Split and Train-Only Preprocessing

The data are split into exactly 201 training observations and 51 held-out test observations using a reproducible 80/20 split. Inspection before splitting found zero missing values, so no imputation or row deletion was required; if a future file contains missing values, its handling must be learned from the training data only.

The exact fixed random seed used for this split was `240201`.

Table 2: Summary Statistics of Training Data

	Variable	Mean	SD	Min	Median	Max
1	brozek	18.67	7.78	0.00	19.00	38.20
2	age	44.89	12.89	22.00	44.00	74.00
3	weight	178.25	29.68	118.50	176.75	363.15
4	height	70.24	2.65	64.00	70.00	77.75
5	adipos	25.39	3.56	18.10	25.00	48.90
6	neck	38.02	2.43	31.10	38.00	51.20
7	chest	100.74	8.38	79.30	99.70	136.20
8	abdom	92.40	10.73	69.40	90.80	148.10
9	hip	99.78	7.10	85.00	99.30	147.70
10	thigh	59.28	5.21	47.20	59.00	87.30
11	knee	38.55	2.44	33.00	38.50	49.10
12	ankle	23.09	1.78	19.10	22.80	33.90
13	biceps	32.23	3.04	24.80	32.00	45.00
14	forearm	28.68	2.05	21.00	28.80	34.90
15	wrist	18.23	0.95	15.80	18.30	21.40

Standardisation. All predictors are centred and scaled using the *training-set* mean and standard deviation. The same transformation is then applied to the test set to prevent information leakage from the test data into the modelling pipeline. Formally, for each predictor j:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j^{\text{train}}}{s_j^{\text{train}}}, \qquad i = 1, \ldots, n.$$

Cross-validation. We use $K = 5$-fold cross-validation (CV) throughout. With CV seed `240301`, one `foldid` is assigned on the 201 training rows (fold sizes 41, 40, 40, 40, and 40) and reused for OLS, Ridge, Lasso, every Elastic Net α, and every random-feature model.

To strictly prevent data leakage, the test set labels are completely held out from model training and tuning. All standardisation is computed exclusively on the training set. We do not standardise within each CV fold since the primary pipeline standardises the entire training block prior to CV, which ensures the same preprocessing statistics are applied consistently across all folds.

1.3 Training-Data Exploration

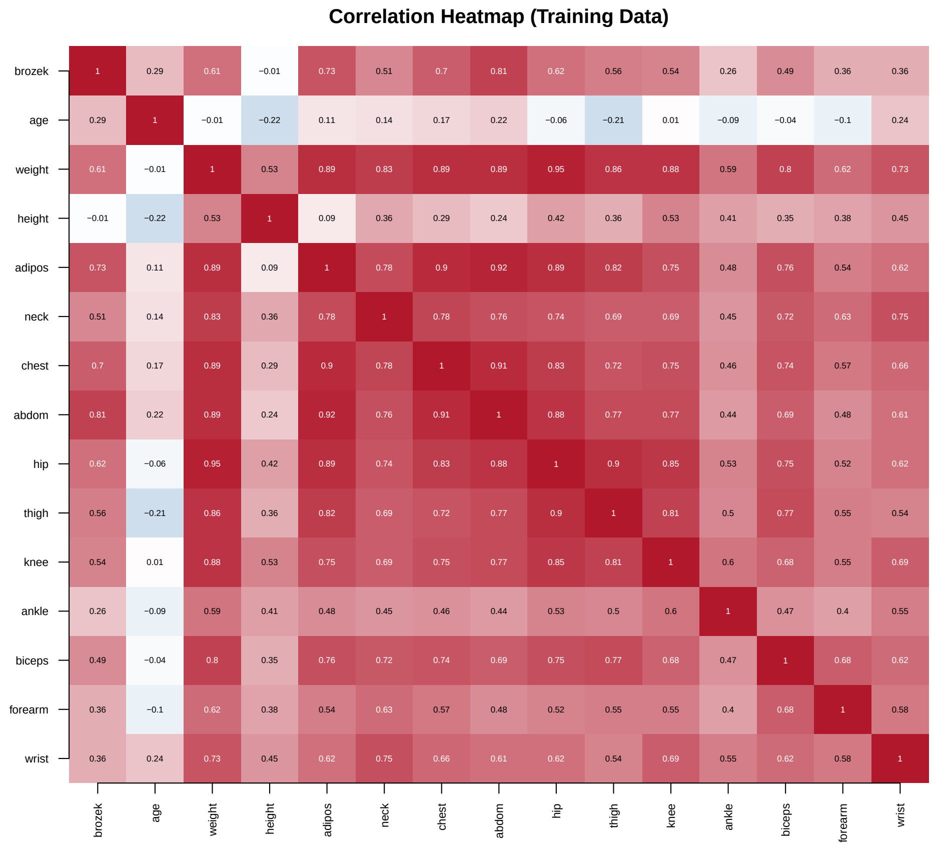

Figure 1: Pearson correlation heatmap of training-set predictors and response.

The heatmap reveals that several anthropometric measurements are highly correlated. For example, training correlations are 0.945 between `hip` and `weight` and 0.894 between `chest` and `weight`; `abdom` also has correlation 0.811 with `brozek`. Such collinearity can make OLS coefficients unstable and motivates regularisation.

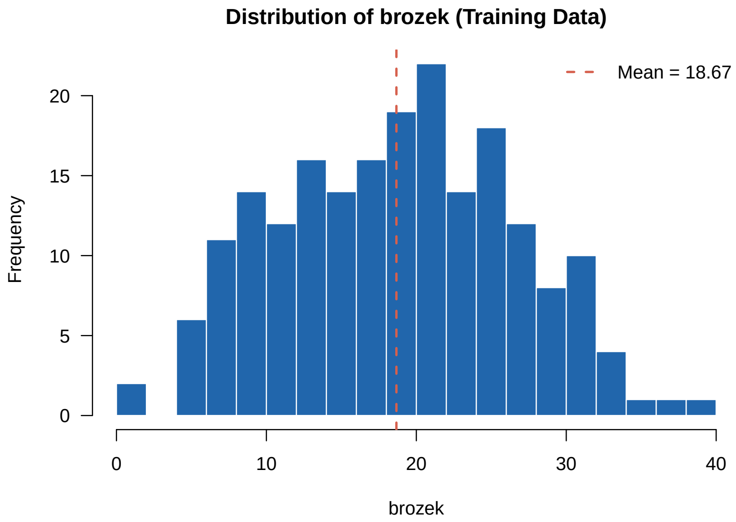

Figure 2: Distribution of the response variable `brozek` in the training set.

The distribution of the response variable `brozek` in the training set is roughly symmetric and bell-shaped, peaking around the mean of 18.67%. It does not exhibit severe skewness, suggesting that no transformation of the response is strictly necessary for linear modelling.

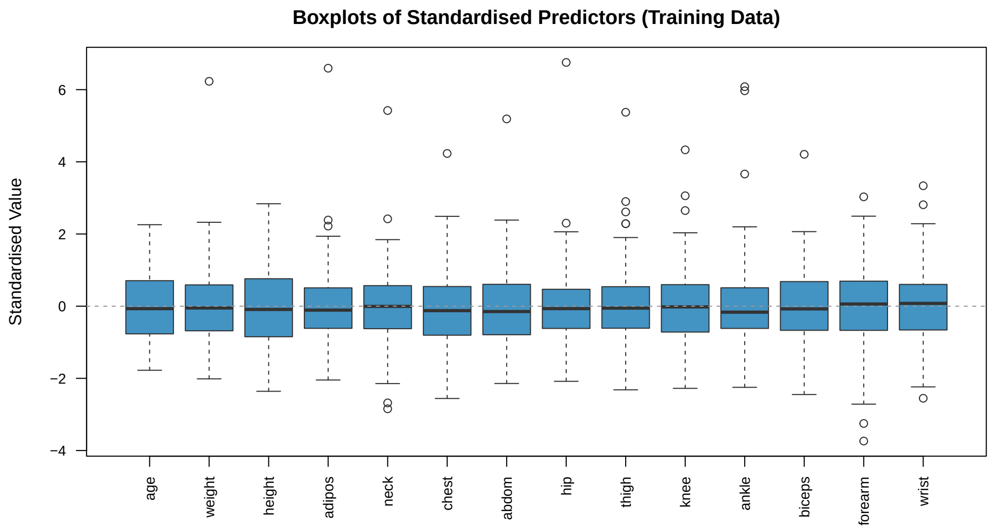

Figure 3: Side-by-side boxplots of standardised predictors (training set).

The side-by-side boxplots show several possible anomalies. In particular, original row 39 is in the training set and has `hip=147.7`, about 6.75 training standard deviations above the training mean. It is retained rather than silently removed, so its possible leverage is visible in the subsequent diagnostics and stability analysis.

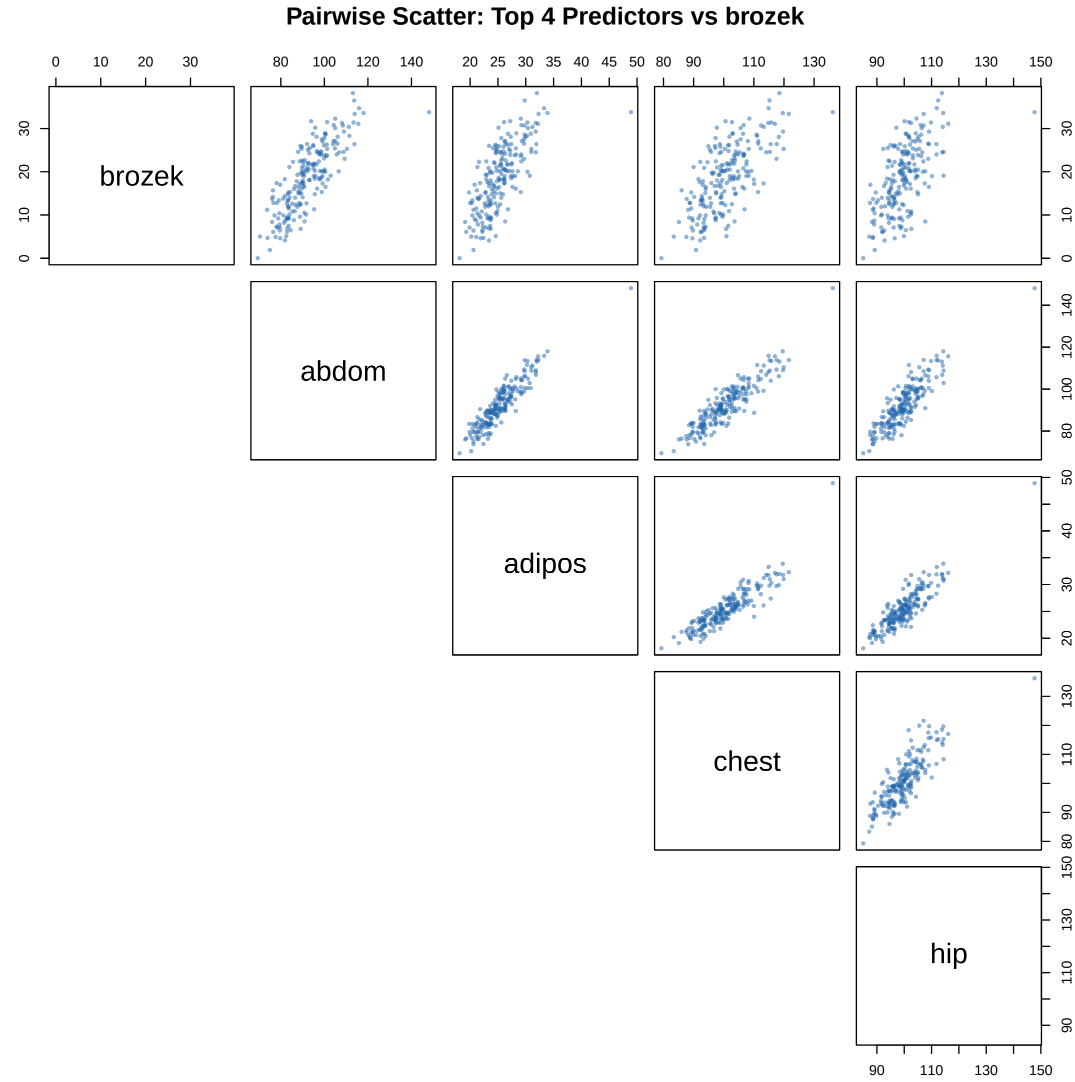

Figure 4: Pairwise scatter plots of selected predictors against `brozek`.

The pairwise scatter plots show a generally linear and positive association between the response and the top four correlated predictors (`abdom`, `adipos`, `chest`, and `hip`). The relationships appear linear without obvious curvilinear patterns, supporting the use of linear models without polynomial transformations. The substantial multicollinearity between these top predictors strongly motivates the use of penalised models like Ridge and Lasso to regularise their coefficient estimates.

Since several body measurements are highly correlated, can regularised regression improve prediction stability and identify which measurements are most useful for predicting `brozek`?

2 OLS, Ridge, and Lasso Regression

2.1 OLS Baseline (P2 — Tuấn)

Ordinary Least Squares provides the unpenalised baseline:

$$\hat{\boldsymbol{\beta}}^{\mathrm{OLS}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 .$$

Table 3: OLS Regression Coefficients (Standardised Predictors)

	Predictor	Coefficient
1	(Intercept)	18.67
2	age	0.79
3	weight	-1.33
4	height	-1.13
5	adipos	-1.32
6	neck	-0.87
7	chest	-0.08
8	abdom	9.47
9	hip	-1.37
10	thigh	1.25
11	knee	0.27
12	ankle	0.26
13	biceps	0.21
14	forearm	1.16
15	wrist	-1.20

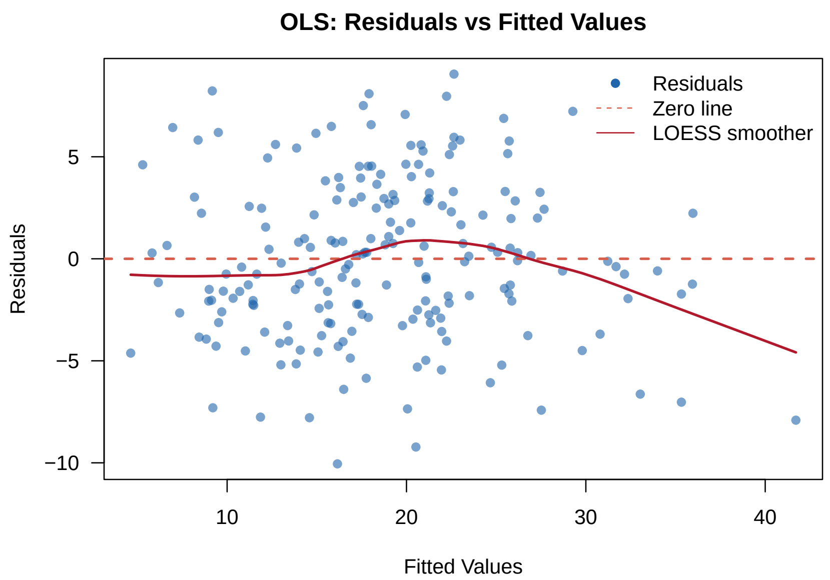

Figure 5: OLS residuals versus fitted values, with a LOESS trend.

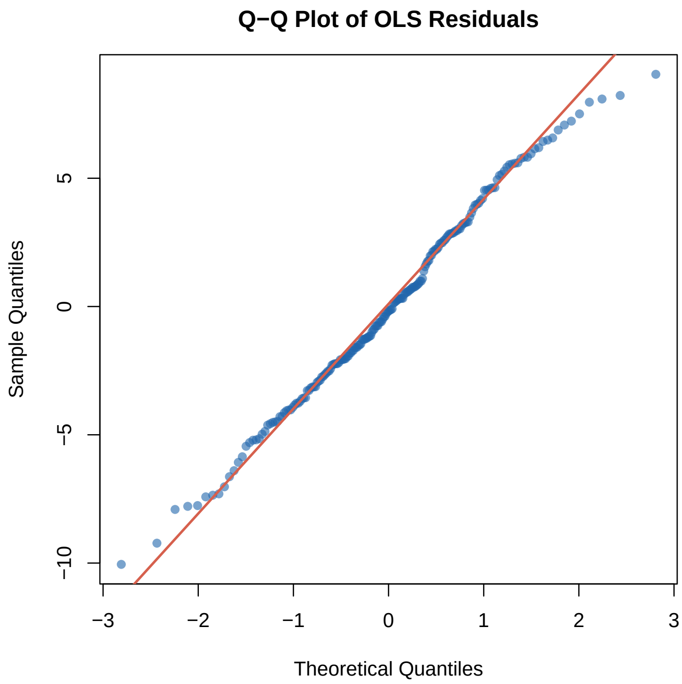

Figure 6: Normal Q-Q plot of the OLS residuals.

With standardised predictors, each slope is the change in body-fat percentage points for a one-training-SD increase, holding the other measurements fixed. `abdom` has the dominant positive slope (9.47), while `weight` (−1.33), `adipos` (−1.32), and `hip` (−1.37) have counterintuitive negative partial slopes. These signs should not be read causally: they occur after conditioning on strongly correlated size measures. OLS has training RMSE 3.884 and shared-fold CV RMSE 4.406. The residual and Q-Q plots show the fit diagnostics without using the holdout data; visible tail departures also caution against relying only on Gaussian-error summaries.

2.2 Ridge Regression (P2 — Tuấn)

Ridge regression adds an ℓ_2 penalty to the OLS objective:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \underset{\boldsymbol{\beta}\in\mathbb{R}^p}{\arg\min} \left\{ \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \frac{\lambda}{2} \|\boldsymbol{\beta}\|_2^2 \right\}.$$

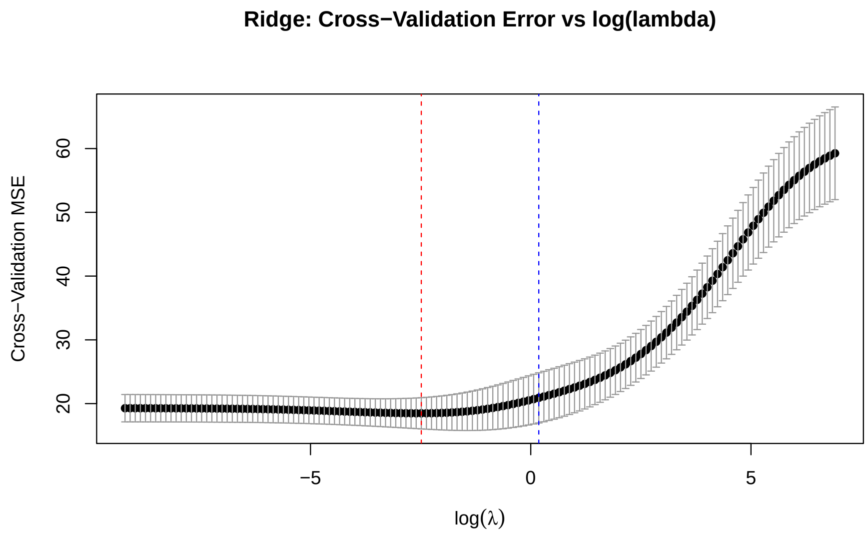

Figure 7: Ridge regression: cross-validated MSE as a function of $\log \lambda$.

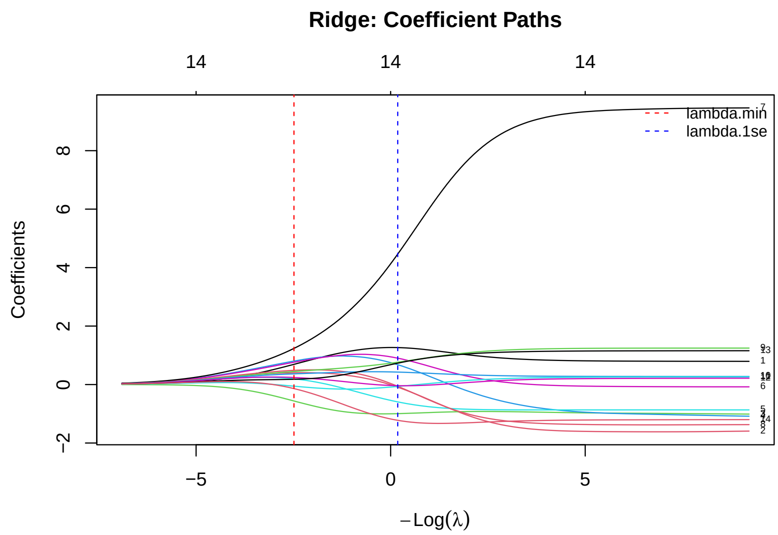

Figure 8: Ridge coefficient path: coefficients vs. $\log \lambda$.

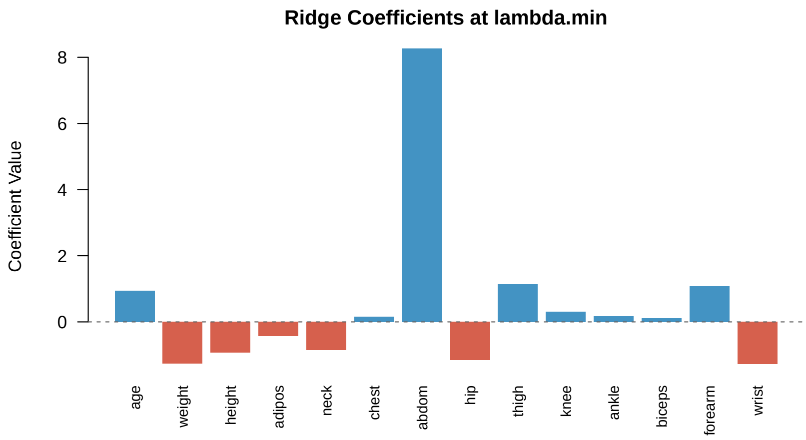

Figure 9: Ridge coefficient estimates at $\lambda_{\min}$ vs. λ_{1se}.

Tuning results. Shared-fold CV selected $\lambda_{\min} = 0.6280$ (CV MSE 19.86, RMSE 4.457) and $\lambda_{1se} = 4.0367$ (CV MSE 23.38, RMSE 4.835). The minimum rule targets the lowest estimated prediction error; the one-SE rule accepts some estimated error for stronger shrinkage.

Table 4: Ridge Regression: Selected Lambda Values

	Metric	Value	CV_Error
1	lambda.min	0.08	18.48
2	lambda.1se	1.20	20.90

Table 5: Ridge Regression Coefficients at $\lambda_{\min}$ and λ_{1se}

	Predictor	Coef_lambda_min	Coef_lambda_1se
1	(Intercept)	18.67	18.67
2	age	0.95	1.26
3	weight	-1.26	0.06
4	height	-0.92	-1.00
5	adipos	-0.43	0.81
6	neck	-0.86	-0.49
7	chest	0.16	0.98
8	abdom	8.26	3.81
9	hip	-1.15	0.13
10	thigh	1.14	0.70
11	knee	0.31	0.44
12	ankle	0.18	-0.10
13	biceps	0.11	-0.02
14	forearm	1.09	0.63
15	wrist	-1.28	-1.12

Ridge retains all 14 slopes but redistributes signal among correlated measurements. For example, `abdom` shrinks from 9.47 under OLS to 5.00 at $\lambda_{\min}$ and 2.15 at λ_{1se}, while correlated size measures receive

Table 6: Condition Numbers: OLS vs Ridge

	Matrix	Condition_Number	Improvement
1	G = X'X/n	3,166.7	—
2	G + lambda*I	104.8	30.2x reduction

smaller, more similar contributions. On the $G = \boldsymbol{X}^\top\boldsymbol{X}/n$ scale, adding $\lambda_{\min}I$ reduces the condition number from 3166.75 to 15.17, a 208.7-fold improvement. No Ridge slope is described as an exact variable-selection result.

2.3 Lasso Regression (P3 — Thuận)

The Lasso replaces the ℓ_2 penalty with an ℓ_1 penalty, enabling automatic feature selection:

$$\hat{\boldsymbol{\beta}}^{\text{Lasso}} = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda\, \|\boldsymbol{\beta}\|_1 \right\}.$$

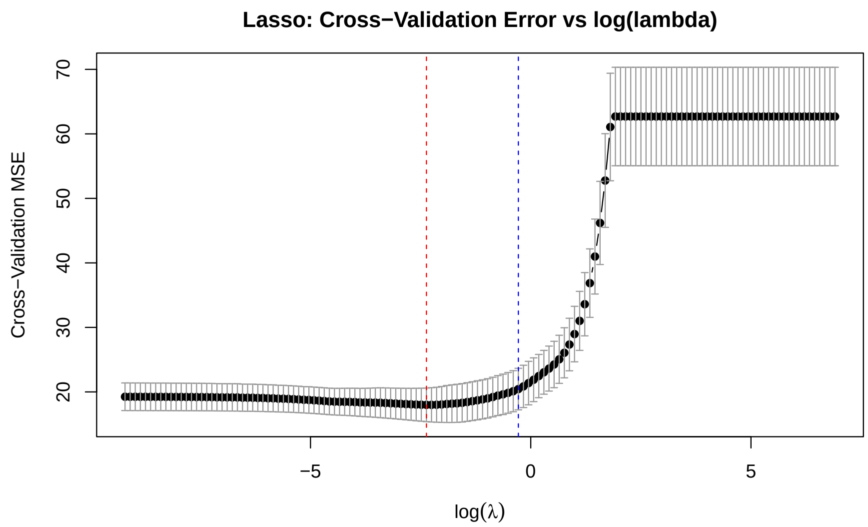

Figure 10: Lasso regression: cross-validated MSE as a function of $\log\lambda$.

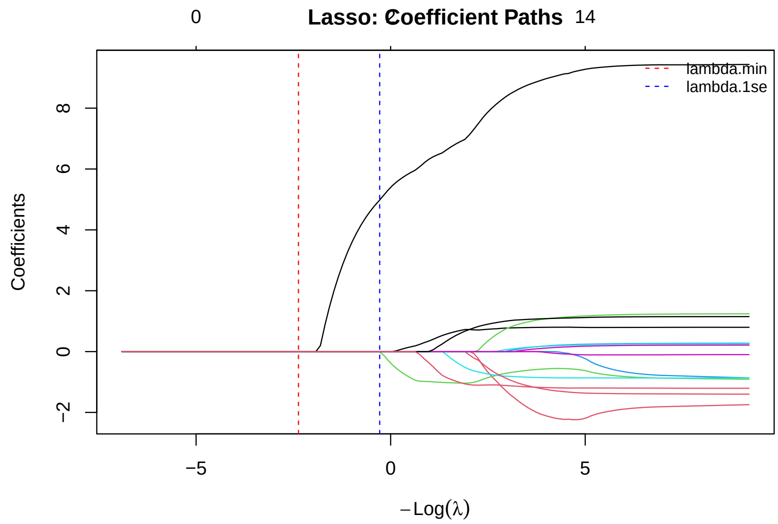

Figure 11: Lasso coefficient path: coefficients vs. $\log\lambda$.

Table 7: Lasso Regression: Selected Lambda Values

	Metric	Value	Nonzero	CV_Error
1	lambda.min	0.09	9	18.00
2	lambda.1se	0.75	3	20.44

Table 8: Lasso Selected Features Under Both Tuning Rules

	Rule	Nonzero	Selected_Features
1	lambda.min	9	age, weight, height, neck, abdom, hip, thigh, forearm, wrist
2	lambda.1se	3	age, height, abdom

At $\lambda_{\min} = 0.0954$, the Lasso CV MSE is 17.99 (RMSE 4.241) and nine slopes are nonzero: `age`, `weight`, `height`, `neck`, `abdom`, `hip`, `thigh`, `forearm`, and `wrist`. At $\lambda_{1se} = 0.7390$, only `age`, `height`, and `abdom` remain and CV MSE rises to 20.39 (RMSE 4.516). Unlike Ridge, the ℓ_1 optimality condition sets other slopes exactly to zero. The selection identifies predictive conditional associations, not causal effects.

Table 9: Lasso Coefficients and Selection at Both Tuning Rules

	Predictor	Coef_lambda_min	Coef_lambda_1se	At_lambda_min	At_lambda_1se
1	(Intercept)	18.67	18.67	Intercept	Intercept
2	age	0.72	0.08	Selected	Selected
3	weight	-0.46	0.00	Selected	Zero
4	height	-0.91	-0.68	Selected	Selected
5	adipos	0.00	0.00	Zero	Zero
6	neck	-0.70	0.00	Selected	Zero
7	chest	0.00	0.00	Zero	Zero
8	abdom	7.69	5.70	Selected	Selected
9	hip	-0.41	0.00	Selected	Zero
10	thigh	0.20	0.00	Selected	Zero
11	knee	0.00	0.00	Zero	Zero
12	ankle	0.00	0.00	Zero	Zero
13	biceps	0.00	0.00	Zero	Zero
14	forearm	0.86	0.00	Selected	Zero
15	wrist	-1.10	0.00	Selected	Zero

Table 10: Comparison of OLS, Ridge, and Lasso Regression

	Model	Tuning	Train_RMSE	Train_MAE	CV_RMSE	L1_Norm	L2_Norm	Nonzero
1	OLS	None	3.88	3.18	4.41	20.73	10.12	14
2	Ridge	lambda.min=0.0833	3.90	3.19	4.30	18.11	8.84	14
3	Lasso	lambda.min=0.0936	3.96	3.20	4.24	13.06	7.96	9

2.4 Training-Based Comparison and Core Model (P3 — Thuận)

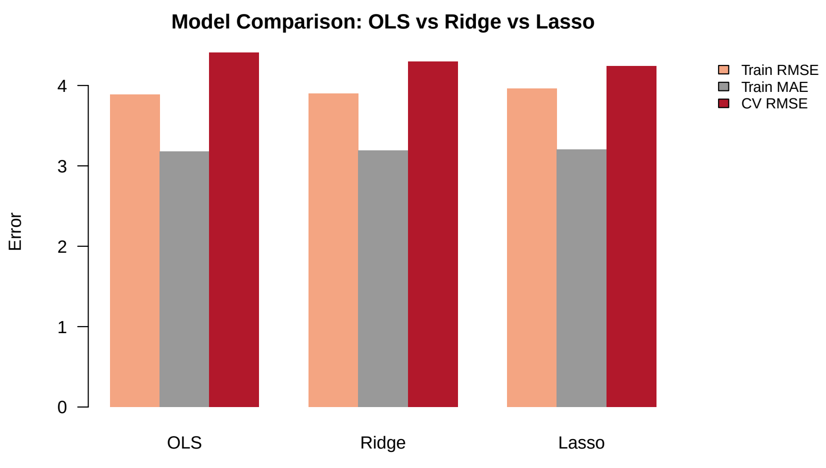

Figure 12: Cross-validated RMSE comparison: OLS vs. Ridge vs. Lasso.

Core Model Declaration (pre-holdout):

Based only on the shared five-fold training comparison, we nominate **Lasso at** $\lambda_{\min} = 0.0954$. It has the lowest CV RMSE (4.241 versus 4.406 for OLS and 4.457 for Ridge) and reduces the model from 14 to nine nonzero slopes. This declaration precedes all holdout scoring.

The $\lambda_{\min}$ rule answers which candidate minimises estimated prediction error, whereas $\lambda_{1\text{se}}$ asks for the most strongly regularised model statistically close to that minimum. Sparsity improves deployment and communication but is not itself evidence of better prediction; the CV error and active-set size therefore appear separately in the comparison.

2.5 Perturbation or Stability Check (P3 — Thuận)

Prediction before check. Before resampling, we predicted that Ridge coefficients would vary less than OLS coefficients, while Lasso would show selection instability among highly correlated measurements.

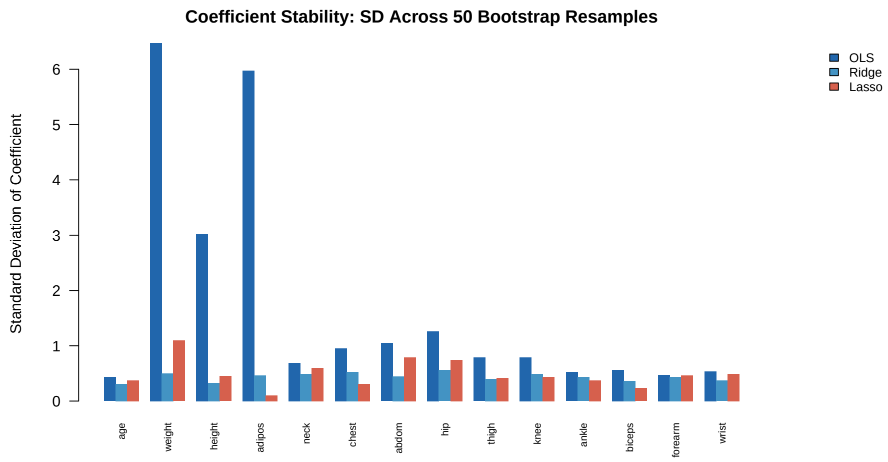

Figure 13: Coefficient stability under predictor perturbation or bootstrap resampling.

Using bootstrap seed 240302, we drew $B = 50$ training resamples and refit the methods without consulting the holdout set. The plot reports slope standard deviations, and the Lasso table records selection frequencies. Ridge is markedly more stable for the most collinear size variables: the bootstrap SD for `weight` is 0.50 under Ridge versus 6.47 under OLS and 1.08 under Lasso. Lasso consistently selects `abdom` and `wrist` (frequency 1.00) and `forearm` (0.98), but alternates among `weight` (0.46), `hip` (0.48), and `chest` (0.26), as predicted for correlated predictors.

3 Mathematical Mechanisms

3.1 Ridge Regression and Conditioning

Let $\boldsymbol{X} \in \mathbb{R}^{n\times p}$ be the centred and scaled training design, let $\boldsymbol{y} \in \mathbb{R}^n$ be the centred training response, and define $G = \boldsymbol{X}^\top \boldsymbol{X}/n$ and $\boldsymbol{z} = \boldsymbol{X}^\top \boldsymbol{y}/n$. The intercept is handled separately and is not penalised. Under

Table 11: Lasso Bootstrap Stability (B = 50)

	Predictor	Selection_Frequency	Coefficient_SD
1	abdom	1.00	0.79
2	wrist	1.00	0.49
3	forearm	0.98	0.47
4	age	0.96	0.37
5	height	0.96	0.45
6	neck	0.82	0.60
7	thigh	0.68	0.42
8	ankle	0.68	0.37
9	biceps	0.50	0.23
10	hip	0.48	0.74
11	weight	0.46	1.09
12	knee	0.46	0.44
13	chest	0.26	0.31
14	adipos	0.12	0.10

the same scaling used by `glmnet`, the Ridge objective is

$$Q_\lambda(\boldsymbol{\beta}) = \frac{1}{2n}\,\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \frac{\lambda}{2}\,\|\boldsymbol{\beta}\|_2^2, \qquad \lambda > 0. \tag{1}$$

Its gradient is

$$\nabla Q_\lambda(\boldsymbol{\beta}) = -\frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) + \lambda\boldsymbol{\beta} \tag{2}$$

$$= G\boldsymbol{\beta} - \boldsymbol{z} + \lambda\boldsymbol{\beta}. \tag{3}$$

Setting this equal to zero gives the normal equations and solution

$$(G + \lambda\boldsymbol{I}_p)\hat{\boldsymbol{\beta}} = \boldsymbol{z}, \qquad \boxed{\hat{\boldsymbol{\beta}}_\lambda^{\mathrm{Ridge}} = (G + \lambda\boldsymbol{I}_p)^{-1}\boldsymbol{z}}. \tag{4}$$

Because G is positive semidefinite, $G + \lambda I_p$ is positive definite for every $\lambda > 0$, so this minimiser is unique.

If $\mu_{\max} \geq \cdots \geq \mu_{\min} \geq 0$ are the eigenvalues of G, adding Ridge translates every eigenvalue by λ. Hence

$$\kappa_2(G + \lambda\boldsymbol{I}_p) = \frac{\mu_{\max} + \lambda}{\mu_{\min} + \lambda} \leq \kappa_2(G). \tag{5}$$

Geometrically, the objective gains curvature in every eigendirection; numerically, its flattest direction is moved away from zero.

For the training design, $\mu_{\max} = 8.942970$ and $\mu_{\min} = 0.002824$, so $\kappa_2(G) = 3166.75$. At the fitted $\lambda_{\min} = 0.627974$,

$$\frac{8.942970 + 0.627974}{0.002824 + 0.627974} = 15.1727,$$

matching Table 6; the condition number falls by a factor of 208.7.

3.2 Lasso Optimality and Soft Thresholding

The Lasso objective on the same scale is

$$R_\lambda(\boldsymbol{\beta}) = \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 . \tag{6}$$

Because $|\beta_j|$ is not differentiable at zero, coordinate j satisfies

$$-\frac{1}{n}\boldsymbol{x}_j^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) + \lambda\,\partial|\hat{\beta}_j| \ni 0, \tag{7}$$

where $\partial|b| = \{\mathrm{sign}(b)\}$ for $b \neq 0$ and $\partial|0| = [-1, 1]$. Writing $r = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}$, this is equivalent to

$$\begin{cases} n^{-1}\boldsymbol{x}_j^\top r = \lambda\,\mathrm{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ \left|n^{-1}\boldsymbol{x}_j^\top r\right| \leq \lambda, & \hat{\beta}_j = 0. \end{cases}$$

When $G = I_p$, the coordinates decouple. Completing the square for coordinate j gives the soft-thresholding rule

$$\boxed{\hat{\beta}_j = \mathrm{sign}(z_j)(|z_j| - \lambda)_+}. \tag{8}$$

Thus $|z_j| \leq \lambda$ produces an exact zero, while larger values are shrunk toward zero by λ. The ℓ_1 constraint has corners on coordinate axes, so a smooth loss contour can first touch a corner. The ℓ_2 constraint is smooth and generally shrinks coefficients without landing exactly on an axis.

3.3 Connecting Algebra to Evidence

The fitted `chest` slope illustrates the different mechanisms. OLS estimates -0.08; Ridge at $\lambda_{\min}$ redistributes signal among the correlated measurements and returns 0.82 but does not select the variable away; Lasso returns exactly zero. The Lasso residual correlation for this coordinate can satisfy the interval condition in Equation 7, whereas Ridge must satisfy the smooth system in Equation 4. Together with the observed condition-number reduction, this connects the algebra to the numerical evidence in Section 2.

4 Elastic Net and Neural-Feature Bridge

4.1 Research Question and Source Map

Research question. When does an ℓ_2 penalty on neural-network weights behave like classical Ridge regression, and why must that connection be qualified for adaptive optimisation and distinguished from dropout? The first source below is a primary research paper not cited in the project brief.

Table 12: Source map linking methodological claims to primary evidence.

Claim	Exact Source and Locator	What It Establishes	What It Does Not Establish
A quadratic weight penalty yields a decay term in gradient learning.	Krogh and Hertz [1991], Section 1, Equations (1)–(2).	Adds $\lambda \sum_i w_i^2/2$ to a neural-network loss and derives the $-\lambda w_i$ update term.	Does not analyse Adam or prove that every modern "weight decay" implementation is equivalent.
ℓ_2 regularisation and weight decay separate under adaptive updates.	Loshchilov and Hutter [2019], abstract and Section 2.	Shows equivalence for standard SGD after learning-rate rescaling and inequivalence for adaptive gradient methods; motivates decoupled AdamW.	Does not make the present frozen-feature experiment a trained neural network.
Dropout randomly masks units during training and uses the unthinned network at test time.	Srivastava et al. [2014], Section 2 and Figure 2.	Defines stochastic activation masking and the corresponding test-time scaling.	Does not establish permanent sparsity or deterministic pruning of learned weights.

4.2 Elastic Net on Original Predictors

The Elastic Net [Zou and Hastie, 2005] combines ℓ_1 and ℓ_2 penalties:

$$\hat{\boldsymbol{\beta}}^{\text{ENet}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right] \right\}.$$

Using the shared folds, λ was tuned separately for $\alpha \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. The pair with the smallest CV MSE was $\alpha = 0.90$, $\lambda_{\min} = 0.1061$, with CV RMSE 4.242 and nine nonzero slopes.

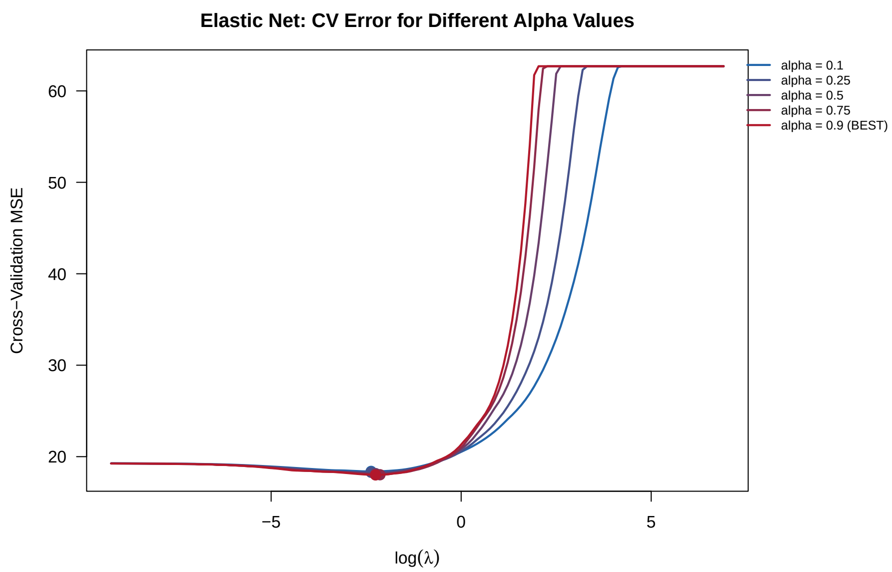

Figure 14: Elastic Net shared-fold CV performance across candidate α values.

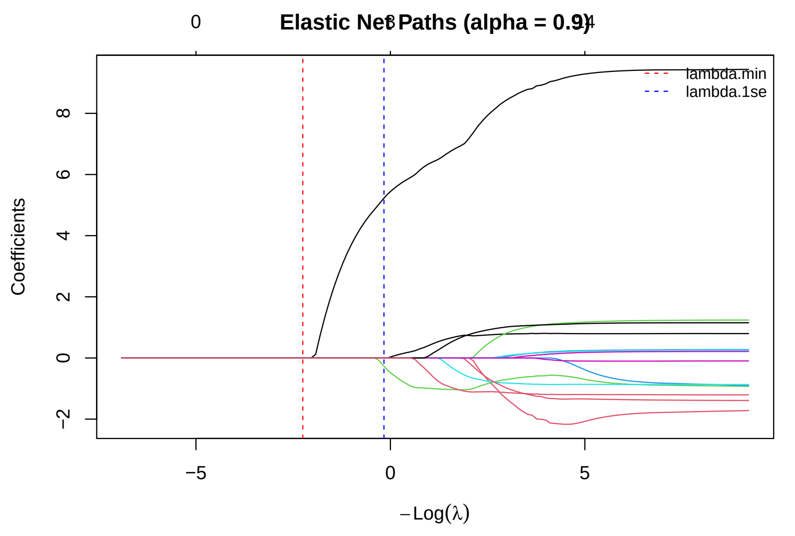

Figure 15: Elastic Net coefficient path for the selected $\alpha = 0.90$.

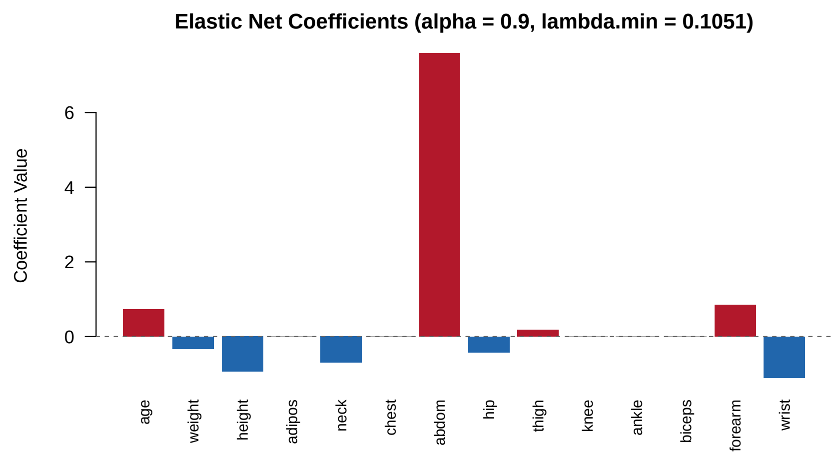

Figure 16: Elastic Net coefficients at the selected (α, λ).

Table 13: Elastic Net: CV Performance Across Alpha Values

	Alpha	Lambda_min	CV_MSE	CV_RMSE	Nonzero
1	0.10	0.09	18.39	4.29	14
2	0.25	0.09	18.29	4.28	12
3	0.50	0.11	18.15	4.26	11
4	0.75	0.12	18.02	4.25	9
5	0.90	0.11	18.00	4.24	9

The selected α lies close to Lasso, and its active set and CV error are accordingly similar. Its nonzero ℓ_2 component still shares shrinkage across correlated predictors, while the dominant ℓ_1 component preserves sparsity; it therefore lies between Ridge's dense solution and Lasso's exact thresholding.

Table 14: Elastic Net Coefficients at Best Alpha = 0.9

	Predictor	Coefficient	Status
1	(Intercept)	18.67	—
2	age	0.73	Active
3	weight	-0.33	Active
4	height	-0.94	Active
5	adipos	0.00	Zero
6	neck	-0.70	Active
7	chest	0.00	Zero
8	abdom	7.59	Active
9	hip	-0.42	Active
10	thigh	0.18	Active
11	knee	0.00	Zero
12	ankle	0.00	Zero
13	biceps	0.00	Zero
14	forearm	0.85	Active
15	wrist	-1.11	Active

4.3 Fixed ReLU Features

With random-feature seed 240401, the standardised predictors are mapped once to $M = 200$ features

$$h_m(\boldsymbol{x}) = \max\{\boldsymbol{a}_m^\top \boldsymbol{x} + c_m, 0\}, \qquad a_{jm} \sim N(0, 1/p), \quad c_m \sim N(0, 0.5^2).$$

The same sampled matrix A and bias vector $\boldsymbol{c}$ are applied to training and test predictors. One constant hidden feature is detected from H_{train} alone and removed from both matrices; centring and scaling of the remaining hidden features are also learned on training rows only. The random hidden weights and biases are fixed. Only the output intercept and slopes are fitted, with λ (and the preselected candidate α values) chosen by the shared training folds.

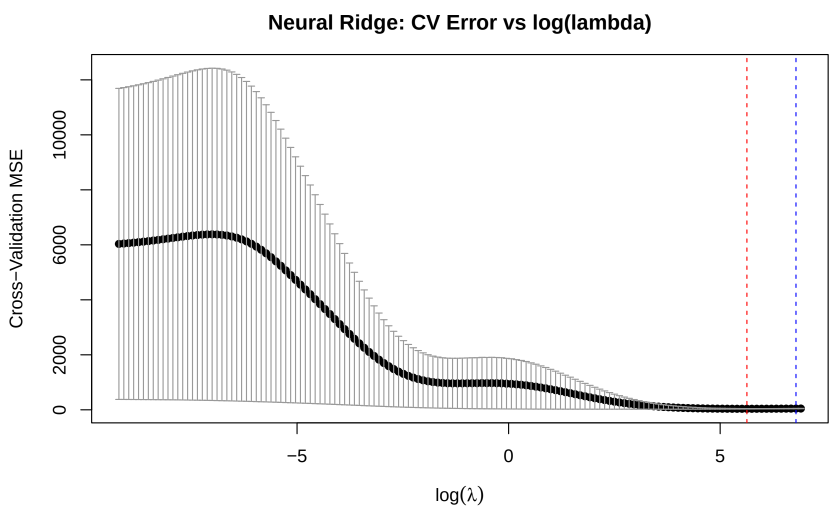

Figure 17: Shared-fold CV curve for Ridge on the fixed ReLU features.

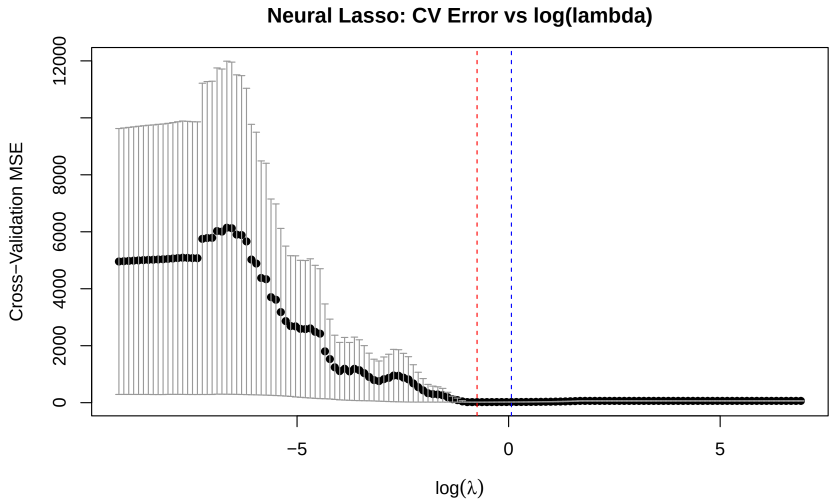

Figure 18: Shared-fold CV curve for Lasso on the fixed ReLU features.

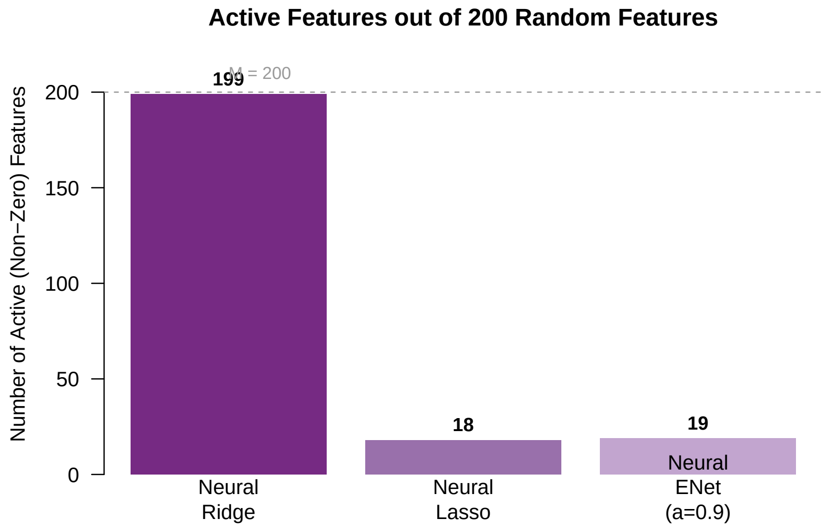

Figure 19: Active output weights for the three fixed-feature models.

Table 15: Neural Random Features: Model Summary (M = 200)

	Model	Alpha	Lambda_min	Train_RMSE	Train_MAE	CV_RMSE	Active_Features
1	Neural Ridge	0.00	279.28	5.51	4.56	6.08	199
2	Neural Lasso	1.00	0.47	4.02	3.24	4.46	18
3	Neural ENet (a=0.9)	0.90	0.47	3.99	3.22	4.46	19

The nonlinear expansion does not improve shared-fold prediction. Neural Ridge has CV RMSE 4.784 with all 199 usable features active; Neural Lasso and Neural Elastic Net reduce this to about 4.390 with 31 active features, but remain worse than the original-predictor Lasso and Elastic Net (about 4.242). With only 201 training cases, the 199-dimensional random representation adds variance and depends on a single frozen draw rather than learning task-specific features.

4.4 Locked Declaration and Final Holdout

Locked Analysis Declaration (before any `y_test` metric)

Training-only centring/scaling is applied unchanged to test predictors. Candidate specifications are OLS; Ridge at $\lambda_{\min} = 0.627974$; Lasso at $\lambda_{\min} = 0.095447$; Elastic Net at $\alpha = 0.90$ and $\lambda_{\min} = 0.106052$; and the fixed-feature models using seed 240401 and $M = 200$. Based on training CV, the deployment model is the **original-predictor Lasso**. The locked primary metrics are RMSE and MAE, with R^2 supplementary.

Table 16: Final Holdout Test-Set Performance (Sorted by RMSE)

	Model	Test_RMSE	Test_MAE	Test_R2	Predeclared
1	Lasso	4.25	3.52	0.68	TRUE
2	Elastic Net (a=0.9)	4.27	3.53	0.68	FALSE
3	Ridge	4.30	3.52	0.68	FALSE
4	OLS	4.48	3.60	0.65	FALSE
5	Neural Lasso	4.52	3.58	0.64	FALSE
6	Neural ENet (a=0.9)	4.59	3.63	0.63	FALSE
7	Neural Ridge	5.31	4.36	0.51	FALSE

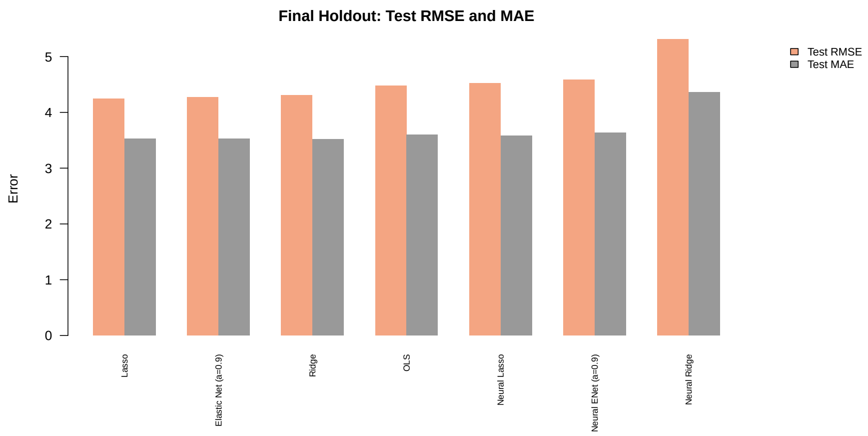

Figure 20: RMSE and MAE for all candidates in the one final holdout evaluation.

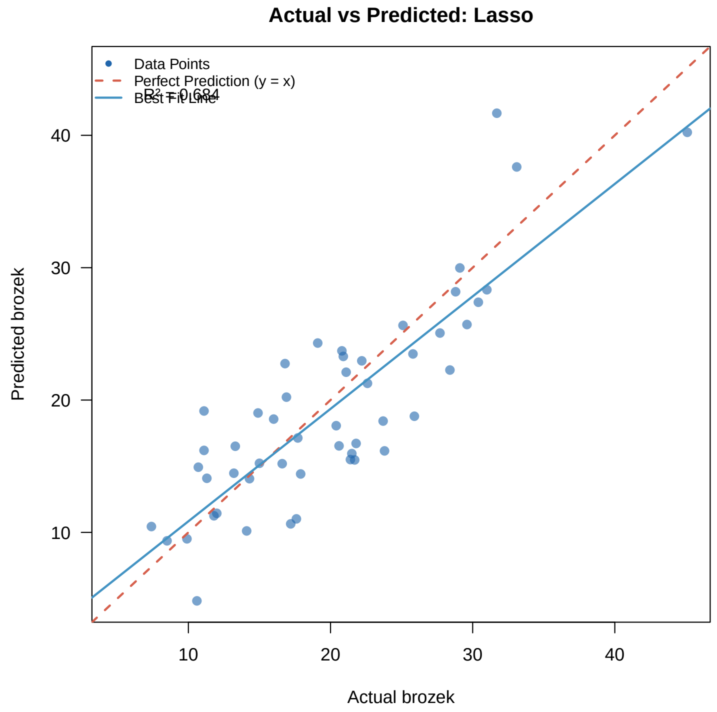

Figure 21: Actual versus predicted `brozek` for the predeclared Lasso on the holdout set.

The holdout evidence supports the declaration: Lasso ranks first with RMSE 4.251, MAE 3.524, and $R^2 = 0.683$. Elastic Net is close (RMSE 4.274), but no model was retuned or replaced after viewing these results. The frozen ReLU models rank below every original-predictor candidate; their best holdout RMSE is 4.757 for Neural Ridge.

4.5 Deep-Learning Connection and Limitations

For a fixed matrix H, an ℓ_2 penalty on the learned output slopes is exactly Ridge regression. In an adaptively optimised neural network, however, adding an ℓ_2 term to the loss need not equal multiplicatively decaying the weights; the update geometry depends on the optimiser [Loshchilov and Hutter, 2019]. A zero output slope here removes one fixed hidden feature from this scalar predictor, but it does not make the input-to-hidden matrix A sparse. Dropout instead samples temporary activation masks during training and uses a scaled unmasked network at test time [Srivastava et al., 2014]; this is not permanent pruning. Most importantly, our single hidden layer is frozen, so the experiment does not establish a result about end-to-end training or a fully trained deep network.

The experiment has at least three limitations. First, it uses only one random feature seed, so the apparent ranking may change with another draw. Second, $n_{\text{train}} = 201$ is small relative to 200 proposed hidden features, making a flexible nonlinear representation difficult to estimate reliably. Third, the hidden weights are not learned and only one width and bias scale are studied. These restrictions make the result a controlled regularisation comparison, not a general assessment of neural networks.

5 Report Quality and Reproducibility

5.1 Reproduction Record

The analysis requires R (the verified run used R 4.5.2). Its direct non-base dependencies, including `glmnet`, `xtable`, `jsonlite`, and `rlang`, and all transitive dependencies are version-pinned in `renv.lock`. The tracked `.Rprofile` points R to the restored project library, which can be rebuilt with `renv::restore()`. A MiKTeX or TeX Live installation must provide `latexmk`, XeLaTeX, and the packages listed in `report/latex-package` pdfLaTeX is not a supported engine because the report contains Unicode Vietnamese names.

From the project root, the following commands restore the exact locked package versions, run every analysis stage in dependency order, record `sessionInfo()`, and build the report.

```
Rscript -e "if (!requireNamespace('renv', quietly = TRUE)) install.packages('renv', repos = 'https://cloud.r-project.org'); renv::restore(prompt = FALSE)"
Rscript R_models/00_run_all.R
cd report
latexmk -C main.tex
latexmk -xelatex -interaction=nonstopmode -halt-on-error main.tex
```

Listing 1: Exact reproduction commands.

The runner executes, in order: `01_data_prep_eda.R`, `02_ols.R`, `02_ridge.R`, `02_lasso.R`, `04_enet.R`, `04_neural.R`, `02_comparison.R`, and `04_holdout.R`. All paths are relative to the project root.

The split seed is 240201, the shared five-fold seed is 240301, the fixed-feature seed is 240401, and the bootstrap stability seed is 240302. The data-preparation script creates and stores the holdout response, but no fitting, tuning, feature-selection, or diagnostic expression references `y_test`. Its first scoring use is in `04_holdout.R`, after the locked declaration has been written; all candidate holdout predictions are then scored together.

The recorded software environment is included verbatim below.

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8

time zone: Asia/Saigon
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] xtable_1.8-8 glmnet_5.0   Matrix_1.7-4
```

```

loaded via a namespace (and not attached):
 [1] compiler_4.5.2   cli_3.6.6        tools_4.5.2      survival_3.8-3
 [5] Rcpp_1.1.2       splines_4.5.2    codetools_0.2-20 grid_4.5.2
 [9] iterators_1.0.14 foreach_1.5.2    jsonlite_2.0.0   shape_1.4.6.1
[13] rlang_1.3.0      lattice_0.22-7
```

Listing 2: R session information generated by the analysis pipeline.

5.2 Recommendation and Limitations

Recommendation. Use the predeclared original-predictor Lasso at $\lambda_{\min} = 0.095447$ for predicting `brozek` in a comparable adult-male population. It had the lowest shared-fold CV RMSE (4.241) and also the lowest one-time holdout RMSE (4.251) and MAE (3.524). Elastic Net was a close second on the holdout set (RMSE 4.274), so the observed advantage should not be overstated.

Prediction versus interpretation. The recommendation prioritises prediction with a smaller deployment model: Lasso retains nine of 14 slopes. If coefficient stability under correlated measurements were the primary objective, dense Ridge estimates would be less variable in the bootstrap check, although Ridge predicts less accurately here. Elastic Net offers a compromise. None of these conditional coefficients or selection events identifies causal effects or a uniquely "true" set of body measurements.

Limitations.

1. The sample contains only 252 adult men from the source study; performance may not generalise to women, children, other populations, or newer measurement protocols.
2. Anthropometric measurements contain measurement error, and the models impose predominantly linear additive structure. The single frozen ReLU expansion is not an exhaustive nonlinear analysis.
3. Tuning estimates depend on one train/test split and one shared fold assignment. The 51-case holdout yields uncertain differences between Lasso and Elastic Net, while the random-feature result is sensitive to its single seed.

5.3 AI Verification Record

AI assistance is documented in Appendix A. The group remains responsible for explaining and independently checking every submitted item.

Table 17: Verification record for AI-assisted work.

AI-Assisted Item	How Checked	Evidence	Resulting Action
R pipeline and leakage audit	Searched every script for holdout-response references and reran all eight stages from the project root.	Pipeline log, five unique fold IDs, generated model objects, and `locked_analysis.txt`.	Kept `y_test` scoring in `04_holdout.R` only.
Mathematical derivations	Matched the displayed objectives to the documented `glmnet` scaling and recomputed the eigenvalue ratio.	$\mu_{\max} = 8.942970$, $\mu_{\min} = 0.002824$, and fitted condition number 15.1727.	Corrected the Ridge n-scaling and normal equations.
LaTeX and report audit	Compiled with XeLaTeX, checked the log for errors/placeholders, extracted the PDF text, and visually inspected rendered pages.	Final PDF, build log, generated figures/tables, and Unicode author names.	Made XeLaTeX explicit and corrected stale asset references.

References

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Anders Krogh and John A. Hertz. A simple weight decay can improve generalization. *Advances in Neural Information Processing Systems*, 4:950–957, 1991.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15 (56):1929–1958, 2014.

Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005. doi: 10.1111/j.1467-9868.2005.00503.x.

A AI Usage Log

This appendix records the AI-assisted work performed during the preparation of this report. Team members remain responsible for reviewing the code, statistical reasoning, and written conclusions.

#	Date	AI Tool & Prompt Summary	Output Used For	Human Verification
1	2026-07-16	Codex: diagnose `setup.R`, package loading, and pipeline failures	Portable project setup, local R library support, and clearer stage logs	Parsed every R script and ran the complete eight-stage pipeline
2	2026-07-16	Codex: audit the R analysis against the quiz specification	Five-fold CV, model summaries, perturbation analysis, neural features, and locked holdout evaluation	Compared generated artifacts with the specification and checked the final metrics
3	2026-07-16	Codex: repair Unicode typesetting and LaTeX build diagnostics	Author names, table layout, package ordering, and build instructions	Compiled with XeLaTeX and inspected the final log and rendered PDF

Table 18: AI usage log for Group 01.